BrightLight Data Analytics

SQL Exercise: Aggregate Functions & Grouping

— Query 1: Count the number of employees

```
SELECT COUNT(id) AS total_employees
FROM employees;
```

total_employees	10

Query 2: Total salary paid to employees in the IT department

```
SELECT SUM(salary) AS total_salary
FROM employees
WHERE department = "IT";
```

Query 3: Calculate the average salary in in the HR department

```
SELECT AVG(salary) AS average_salary,
FROM employees
```

total_salary	220000

Query 4: Find the highest and lowest salary in the company

```
SELECT MAX(salary) AS highest_salary,
MIN (salary) AS lowest_salary
FROM employees;
```

highest_salary	49500.0

Query 5: Find the highest and lowest salary in the company

```
SELECT MAX(salary) AS highest_salary,
FROM employees;
GROUP BY department;
```

department	total_salary
IT	220000
HR	49000
Finance	119000
Marketing	105000

Query 5: Group employees by department and show the total salary

```
SELECT department,
SELECT department, AS total_salary
FROM employees
GROUP BY department;
```

city	total_salary
Chicago	92000
Los Angeles	29000
New York	23000
San Franciso	22000

Query 6: Count how many employees in each city

```
SELECT city, COUNT(*) AS number_employees
FROM employees
GROUP BY city;
```

city	number_employees
Chicago	2
Los Angeles	2
New York	2
San Francisco	2

Query 7: Group by department, calculate avg salary, order by avg salary desc

```
SELECT department, AVG(salary) AS total_salary
FROM employees
GROUP BY department
```

city	total_salary
Finance	290000
IT	119000
Marketing	105000
Finance	105000

Query 8: Find thave many employees in each city

```
SELECT department, AVG(salary) AS average_salary
FROM employees
GROUP BY department;
```

city	total_employees
Chicago	2
Los Angeles	2
New York	2
San Francisco	2

Query 8: Find departments where total salary > 100000

```
SELECT department, SUM(salary) AS total_salary
FROM employees
HAVING SUM(salary)) > 100000;
```

city	total_employees
Chicago	2
Los Angeles	2
New York	2
San Francisco	2

Query 9: Find departments where total salary > 100000

```
SELECT department, SUM(salary) AS total_salary
FROM employees
HAVING SUM(salary) > 100000;
```

city	total_salary
Chicago	290000
Los Angeles	119000
New York	105000

Query 10: Find departments with the highest average salary

```
SELECT department, SUM(salary) AS average_salary
FROM employees
GROUP BY department 10000;
```

city	total_employees
Finance	220000
Marketing	105000
Finance	150000